

Software Project and Management



Assignment 1 (First Deliverable)

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SE-E

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Agent: Nutrition Coach Agent

1. Project Lifecycle & Mapping with Process Groups:

A. Lifecycle Phases:

i. Initiation Phase (Weeks 1–2)

- Approve the Nutrition Coach Agent as the official project topic and define its scope: an agent that adapts diet plans based on user patterns and goals (weight loss, muscle gain, balance).
- Form a team of three members and assign roles: Project Manager, Developer, and AI/ML Developer.
- Identify stakeholders: instructor, project team, and section integration coordinator.
- Prepare and submit the Project Charter including objectives, milestones, risks, and budget information.

ii. Planning Phase (Weeks 3–4)

- Create a Work Breakdown Structure (WBS) covering major tasks: dataset creation, rule engine development, goal adaptation, integration APIs, testing, reporting.
- Develop a baseline project schedule and Gantt chart with milestones: dataset finalized, prototype built, testing completed, integration achieved.
- Estimate project budget (time and effort cost, no direct monetary cost since synthetic data and free tools will be used).
- Identify project risks such as data inaccuracy, user engagement challenges, limited scope, and integration difficulties.
- Define quality parameters: accuracy of diet suggestions, smooth integration with section application, and timely delivery of milestones.
- Finalize synthetic nutrition datasets (weekly food logs including calories, macros, meal types) and rules for personalized recommendations.

iii. Execution Phase (Weeks 5–10)

- Generate synthetic food log datasets with attributes like calories, protein, carbs, fats, and daily intake patterns.
- Develop the rule-based engine for nutrition recommendations using predefined thresholds and health guidelines.

- Implement goal-based adaptation features: weight loss (calorie deficit), muscle gain (protein surplus), balanced diet (macro stability).
- Build API endpoints for data exchange and integration with the section-wide agentic application.
- Create a working prototype of the Nutrition Coach Agent and test it with sample user goals.

iv. Monitoring & Controlling Phase (Weeks 6–12)

- Track progress against baseline Gantt chart milestones.
- Monitor planned vs. actual effort and budget usage.
- Conduct unit testing and validation of recommendations to ensure they match user goals and dataset logic.
- Identify and resolve issues such as inaccurate outputs, delays, or integration conflicts.
- Apply corrective actions for scope or timeline deviations.
- Conduct weekly review meetings and submit progress reports to the instructor.

v. Closing Phase (Weeks 13–14)

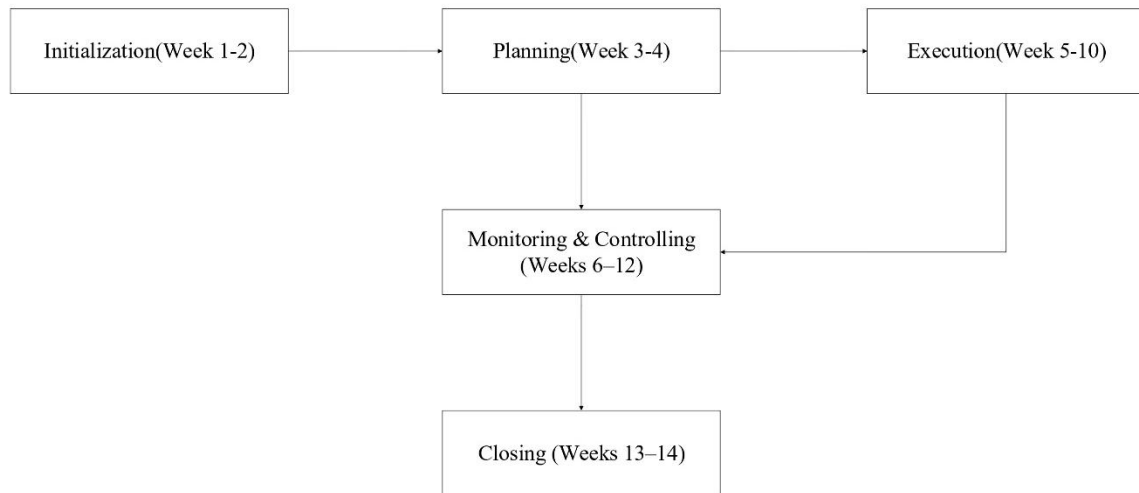
- Perform final integration testing of Nutrition Coach Agent within the section-wide multi-agent system.
- Prepare and deliver the final presentation and live demo, showing recommendations adapting to different user goals.
- Submit the final consolidated project report containing all project management documents (charter, WBS, schedule, cost plan, risk plan, monitoring reports).
- Prepare and submit a Lessons Learned Report reflecting challenges (data accuracy, integration), achievements, and areas of improvement.
- Update and submit all refined project documents (charter, schedule, risk plan, etc.).
- Obtain formal approval and sign-off from instructor/stakeholders to officially close the project.

B. Mapping with Five Process Groups:

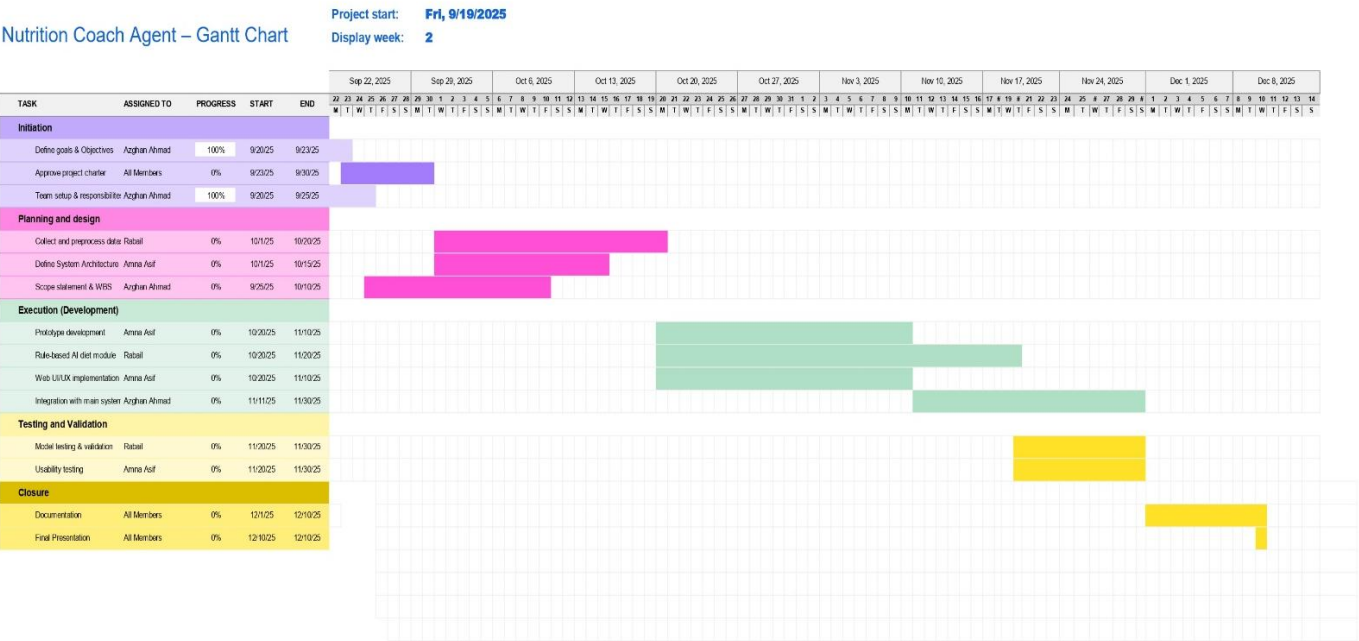
Knowledge Area	Initiating	Planning	Executing	Monitoring & Controlling	Closing
Project Integration Management	Develop project charter	Develop project management plan	Direct & manage project work (dataset creation, prototype design, integration)	Monitor & control project work, Perform integrated change control	Close project, Final presentation & report
Project Scope Management	Define high-level scope	Plan scope management, Collect requirements, Define scope, Create WBS	Execute scope tasks (diet rules, personalized recommendations)	Validate scope, Control scope	Confirm scope completion
Project Time Management	—	Plan schedule management, Define & sequence activities, Estimate resources & durations, Develop schedule	Follow project timeline	Control schedule adherence	—
Project Cost Management	—	Plan cost management, Estimate costs, Determine budget	Track expenses (tools, software, labor)	Control costs	—
Project Quality Management	—	Plan quality management	Perform quality assurance (testing diet suggestions & accuracy)	Control quality	—
Project Human Resource Management	Identify team members & roles	Plan human resource management	Acquire & manage project team, Assign roles	Monitor team performance	Release team members

Project Communications Management	Identify communication needs	Plan communication management	Manage communications (weekly meetings, updates)	Control communications flow	—
Project Risk Management	Identify initial risks	Plan risk management, Analyze & plan responses	Execute risk responses (dataset issues, low engagement)	Monitor & control risks	—
Project Procurement Management	—	Plan procurement (tools, licenses, datasets)	Conduct procurements (if needed)	Control procurements	Close procurements
Project Stakeholder Management	Identify stakeholders (instructor, teammates, users)	Plan stakeholder engagement	Manage stakeholder communication & feedback	Control stakeholder satisfaction	Confirm stakeholder approval

C. Visual Representation of Planned Lifecycle:



2. Baseline Gantt Chart (14 Weeks):

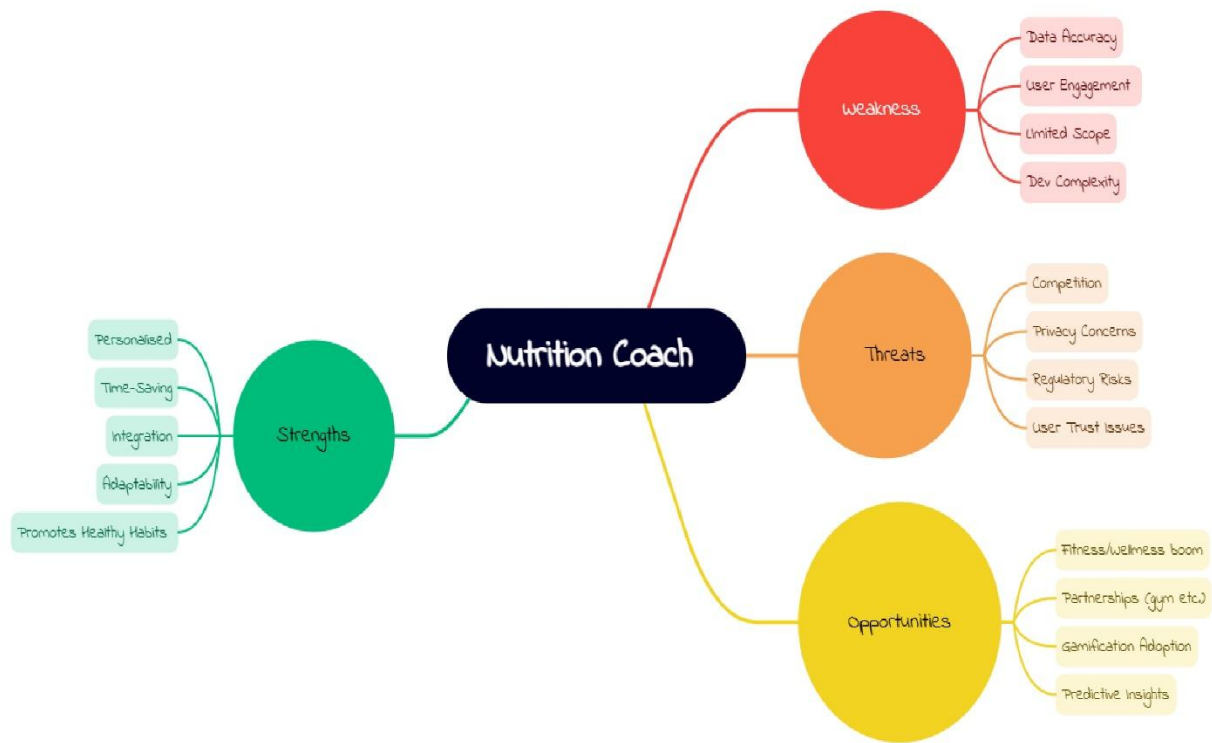


Explanation:

Phase	Task	Assigned To	Start Date	End Date	Duration
Initiation (Weeks 1–2)	Define Goals & Objectives	Azghan Ahmad	9/19/2025	9/23/2025	5 days
	Approve Project Charter	All Members	9/19/2025	9/25/2025	7 days
	Team Setup & Responsibilities	Azghan Ahmad	9/19/2025	9/25/2025	7 days
Planning & Design (Weeks 3–4)	Collect & Preprocess Data	Rabail	10/2/2025	10/9/2025	8 days
	Define System Architecture	Amna Asif	9/25/2025	10/10/2025	16 days
	Scope Statement & WBS	Azghan Ahmad	9/25/2025	10/5/2025	11 days

Execution (Development) (Weeks 5–10)	Prototype Development	Amna Asif	11/1/2025	11/20/2025	20 days
	Rule-Based AI & Food Model	Amna Asif	11/5/2025	11/25/2025	21 days
	Web UI/UX Implementation	Amna Asif	11/3/2025	11/30/2025	28 days
	Integration with Main System	Azghan Ahmad	11/15/2025	11/30/2025	16 days
Testing & Validation (Weeks 11–12)	Model Testing & Validation	Rabail	11/25/2025	12/5/2025	11 days
	Usability Testing	Amna Asif	11/25/2025	12/5/2025	11 days
Closure (Weeks 13–14)	Documentation	All Members	12/5/2025	12/10/2025	6 days
	Final Presentation	All Members	12/10/2025	12/13/2025	4 days

3. SWOT Analysis:



Presented with xmind

a. Strengths:

- **Personalised** → Provides customized diet plans tailored to individual needs.
- **Time-Saving** → Saves users' time by automating meal planning and tracking.
- **Integration** → Can integrate with wearable devices or fitness apps.
- **Adaptability** → Adapts to diverse dietary preferences and restrictions.
- **Promotes Healthy Habits** → Encourages sustainable lifestyle and nutrition choices.

b. Weaknesses:

- **Data Accuracy** → Relies heavily on the accuracy of nutritional databases.
- **User Engagement** → Risk of low long-term engagement if not interactive enough.
- **Limited Scope** → May only cover basic diet tracking instead of holistic wellness.

- **Development Complexity** → Building AI-driven personalized recommendations is technically challenging.

c. Opportunities:

- **Fitness Wearables Boom** → Growing market of smartwatches and health trackers supports integration.
- **Partnerships** → Potential to collaborate with fitness brands, gyms, or diet companies.
- **Gamification Adoption** → Adding challenges and rewards can boost user motivation.
- **Predictive Insights** → Use AI to predict health trends and prevent future issues.

d. Threats:

- **Competition** → Strong competitors already exist in nutrition and fitness apps.
- **Privacy Concerns** → Handling personal health data creates security risks.
- **Regulatory Risks** → Must comply with health and data protection regulations.
- **User Trust Issues** → Users may hesitate to trust AI for sensitive health advice.

4. Project Charter:

Project Charter – Nutrition Coach Agent

Field	Details
Project Title	Nutrition Coach Agent
Date of Authorization	19th September 2025
Project Start Date	20th September 2025
Planned Finish Date	10th December 2025
Key Schedule Milestones	• Approve project charter by September 30, 2025 • Complete dataset generation by October 20, 2025 • Complete prototype development by November 10, 2025 • Complete integration and testing by November 30, 2025 • Final presentation and closing deliverables by December 10, 2025
Budget Information	The university has allocated resources equivalent to a standard semester-long project, with additional support for: • Cloud infrastructure (free student credits where possible) • Research papers, datasets, and development tools (internal academic resources) • No external hiring; all work will be divided among three student developers
Project Manager	Azghan Ahmad , +92-366493156, azghanduplicate786@gmail.com
Project Objectives	The Nutrition Coach Agent project aims to develop a rule-based AI agent that generates and adapts personalized diet plans for users based on their eating patterns and health goals. The project will focus on creating synthetic datasets, implementing adaptive diet rules, and integrating with the section-wide application.
Main Project Success Criteria	• The agent must generate personalized diet recommendations accurately. • The system must integrate successfully into the section-wide application. • The project must be completed within the planned schedule and budget.

	Final deliverables must be well-documented, tested, and formally approved by the instructor.			
Approach	- Divide responsibilities among three core developers to ensure balanced workload. - Within one month, prepare scope statement, work breakdown structure, and Gantt chart for effective tracking. - Collect nutrition datasets and finalize preprocessing pipeline within two months. - Conduct weekly project meetings for progress updates and issue resolution. - Test AI models with real-world data to validate recommendation accuracy before deployment. - Document all phases to ensure reproducibility and future improvements			
Roles & Responsibilities	Name	Role	Position	Contact Information
	Azghan Ahmad	Project Manager	Student Lead	i222667@nu.edu.pk
	Amna Asif	Team Member	Web Developer	i228777@nu.edu.pk
	Rabail	Team Member	AI/ML Engineer	i221507@nu.edu.pk
Comments	<i>"I will focus on ensuring the AI/ML models are accurate and practical for real-world use."</i> – Rabail <i>"I will develop a user-friendly web interface so users can easily interact with the system."</i> – Amna Asif			